



Task: Sensitive Data Classification & Context

The Current Situation & Business Problem

Today, our detection systems rely on strict rules and Regular Expressions (Regex) to scan data and filter out problematic or sensitive parts (e.g., usernames, credit card numbers, encryption keys, etc.).

This method generates a massive amount of False Positives. The reason is that Regex only considers the text pattern and ignores the **context** or the specific location where the secret is found. The result is "Alert Fatigue" for our customers, leading them to ignore the system and creating a real risk of missing actual data leaks.

Your Challenge

Your goal is to make a leap from "dumb," rigid detection to smart, context-based detection. You need to develop a mechanism that can assess whether the identified information is truly sensitive or just a false positive, relying on metadata, location, and a deeper understanding of the data.

Environment & Data

- **Independent Data Creation:** You will not receive a pre-made Data Set. You must generate the Mock Data and the test sets for your system yourselves.
- **Reference Only:** You can look at files and existing types of secrets in the system to understand the data structure and the threats we deal with today, but you cannot use them as a database for the exercise.

Expected Deliverables

At the end of the exercise, you must present a **working software application including a Graphical User Interface (GUI)** that performs the following actions:

- **Input:** Receives a list of processes.
- **Processing:** Analyzes the processes and identifies sensitive data. You can (and should) use Artificial Intelligence (AI) models when there is a high level of certainty, but you must pay close attention to how sensitive data is handled and protected.
- **Output:** Displays the processes while filtering, redacting, or masking the located sensitive parts.



Technology, Permissions & Freedom of Action

- **Total Freedom:** You have absolutely no technological, language, or architectural limitations. Choose the tools that are best suited to solve the problem.
- **Infrastructure & Permissions:** Do you need AWS access? Special permissions? API keys for external models? Cloud environments? Everything is possible. Just reach out to your mentor, and they will make sure you get whatever you need to sprint forward.

Success Metrics

Your system will be measured on the delicate balance between two critical requirements:

- **Minimum False Positives:** A drastic reduction in false alerts compared to the current situation.
- **Zero Missed True Positives:** The system must not miss real secrets. When in doubt, it is better to be strict than to let a secret leak.

Notes & Emphases

- **We do not see the content of the files themselves:** It is important to remember that your system cannot rely on reading the full content of a file. Instead, you must learn to rely on the company's behavioral patterns, the behavior of various software programs within the organization, and the smart use of context and metadata.
- **Viewing real data for learning purposes:** For research and understanding the problem, you are allowed to look at real customer data in the Console under the **Data/Secrets** category. Remember – this data is intended solely to give you an idea of how things look in practice, and must not be used as the database for the exercise.
- **Access to Security Analysts' work:** Upon request, we can grant you access to a group where our analyst team currently reviews alerts one by one and manually disqualifies items that are likely not real secrets. The purpose of granting this access is to give you more insight into what we currently do inefficiently, so you can learn from it and improve the process.

Ideas for Expansion (Bonus/Next Steps)

If everything is running smoothly and you want to take your system a step further, here are some possible directions for expansion:



- **Custom Data Classification:** Build the capability to explore and manage classifications of sensitive data, including a UI that allows for creating "Custom Rules" tailored specifically to the needs, environment, or terminology of a specific organization.
- **Dynamic Risk Scoring:** Move from a binary decision ("secret / not a secret") to a system that assigns a risk score. The score could be weighted by various parameters such as the age of the information, the type of process accessing it, and the environment in which it runs.
- **Process Lineage Investigation:** Integrate context on the process's origin. Was the action performed by a routine automated script (like a known CI/CD pipeline) or by a new, interactive connection (like an unusual SSH login)?
- **Building a Behavioral Baseline:** Use AI to learn what is considered routine (normal) behavior in the organization for specific processes or files, and trigger alerts only when there is a deviation from this norm.
- **Visual Mapping of Connections:** Add a graphical display in the UI that shows the "tree" or chain connecting the suspicious process, the server it runs on, and the information it attempted to access.
- **The Sky is the Limit:** Creative and additional ideas are highly welcome and could very well make it into our final product! You are not limited to the directions suggested here; take it in whatever direction you want, everything is open as long as you are solving the core business problem.

We Are Here For You

Throughout the exercise, your mentor is available to you. Whether you need technical approval, permission unlocks, advice on architecture, or just guidance when you're stuck – don't hesitate to use them. Our goal is for you to succeed in delivering the best possible product.

Good luck,
Yonatan Lewkowicz.